

THE DFIR CHRONICLES



CVE-2026-31431 "Copy Fail"
write 4 controlled bytes
into page-cache of /usr/bin/su
on-disk file: CLEAN
FIM: OK
Only the process tree knows.
00000000 mov eax, 0x699682f
00000001 mov ebx, 0x0000000
00000002 mov ecx, 0x06b732f
00000003 mov edx, 0x0000000

SOC
NIGHT SHIFT

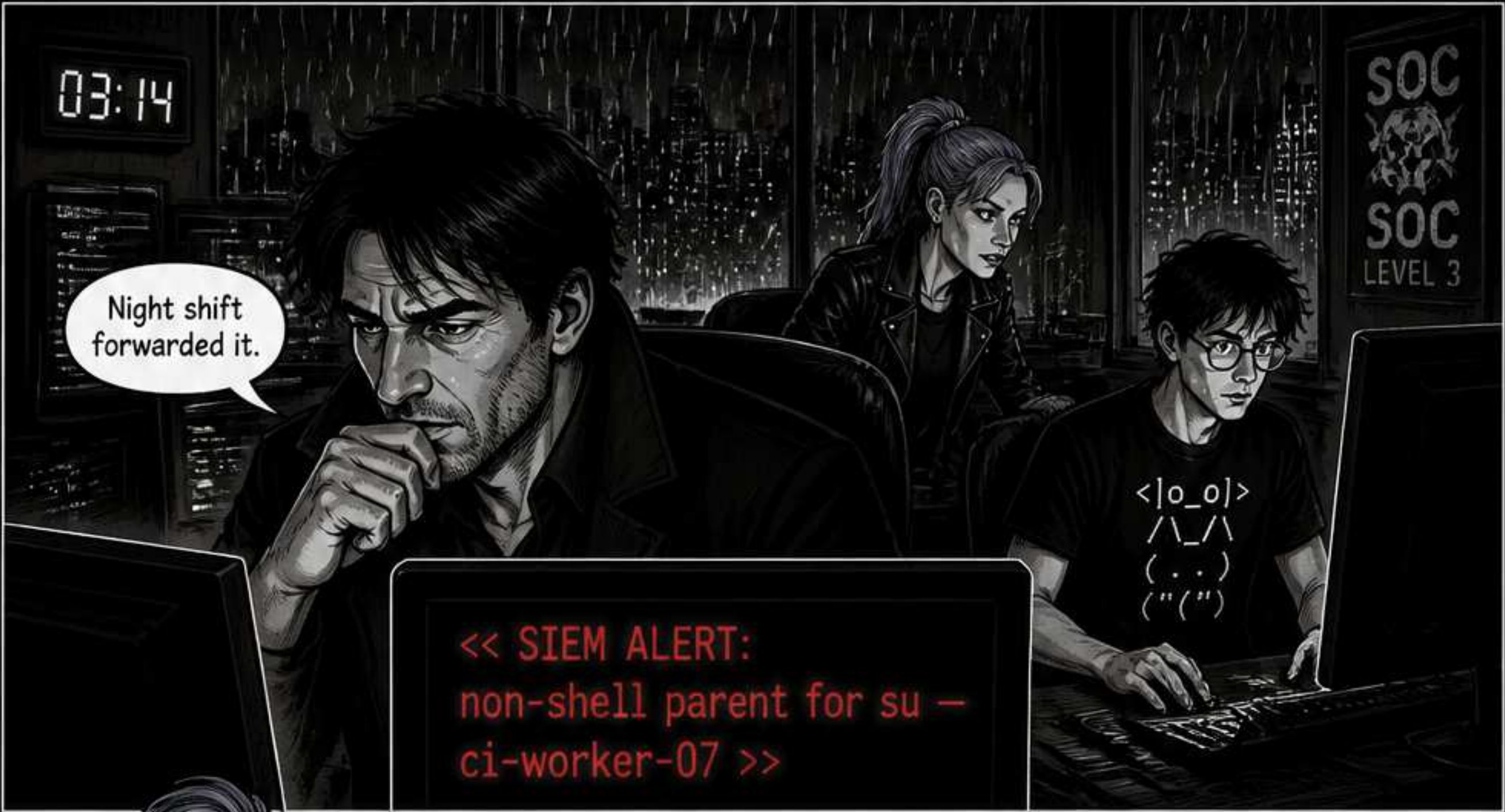
EVERY LOG
TELLS A LIE
EXCEPT THE
PROCESS TREE

(͡° ͜ʖ ͡°)
// bin/bash
sleep(0):
exploit():

03:14:07 [ALERT] non-shell parent for su
host=ci-worker-07 user=devops_leo
src_ip=103.248.7.91 geo=SG
process=su uid=0 outcome=SUCCESS
parent=python3 pid=42411
cmdline=python3 /dev/shm/poc.py

FOUR BYTES TO ROOT

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```
<< Velociraptor: !run Custom.Linux.Sys.OSInfo >>
```

```
Hostname      : ci-worker-07
```

```
OS            : Ubuntu 24.04.2 LTS (Noble Numbat)
```

```
Kernel       : 6.17.0-9-generic
```

```
Architecture : x86_64
```

```
Role         : CI/CD build worker
```

```
=====
```

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OSInfo

=====

Resolving host operating system information...

[*] Querying CI environment inventory...

[*] Correlating host metadata...

[*] OS fingerprint matched.

<< Hostname: ci-worker-07 >>

<< Ubuntu 24.04.2 LTS -- Noble Numbat >>

<< Kernel Release: 6.17.0-9-generic >>

<< Arch: x86_64 -- CI/CD build worker >>

THAT KERNEL
SHIPS ALGIF_AEAD.

01100100 01100100
01001001 01010010
~_(_)/_/
</forensics>

KEEP A KERNEL
LPE ON THE TABLE.

DFIR
FUEL



Who came in?
From where?

```
!run Custom.Linux.Sys.RSyslogSSHLogin >>
```

Time	User	IP	Method	Result
2026-05-21 14:18	marlowe	10.42.1.18	publickey	Accepted
2026-05-21 22:48	dvogel	91.214.63.117	password	Accepted

```
!run Linux.Sys.LastUserLogin >>
```

Time	User	IP	Method	Result
2026-05-21 14:18	marlowe	10.42.1.18	publickey	Accepted
2026-05-21 22:48	dvogel	91.214.63.117	password	Accepted

```
!run Custom.Linux.Sys.RSyslogSSHLogin >>
```

Time	User	IP	Method	Result
2026-05-21 14:18	marlowe	10.42.1.18	publickey	Accepted
2026-05-21 22:48	dvogel	91.214.63.117	password	Accepted

```
!run Linux.Sys.LastUserLogin >>
```

User	From	TTY	Login
marlowe	10.42.1.18	pts/1	2026-05-21 14:18
dvogel	91.214.63.117	pts/0	2026-05-21 22:48

marlowe -
publickey,
internal.

And dvogel?

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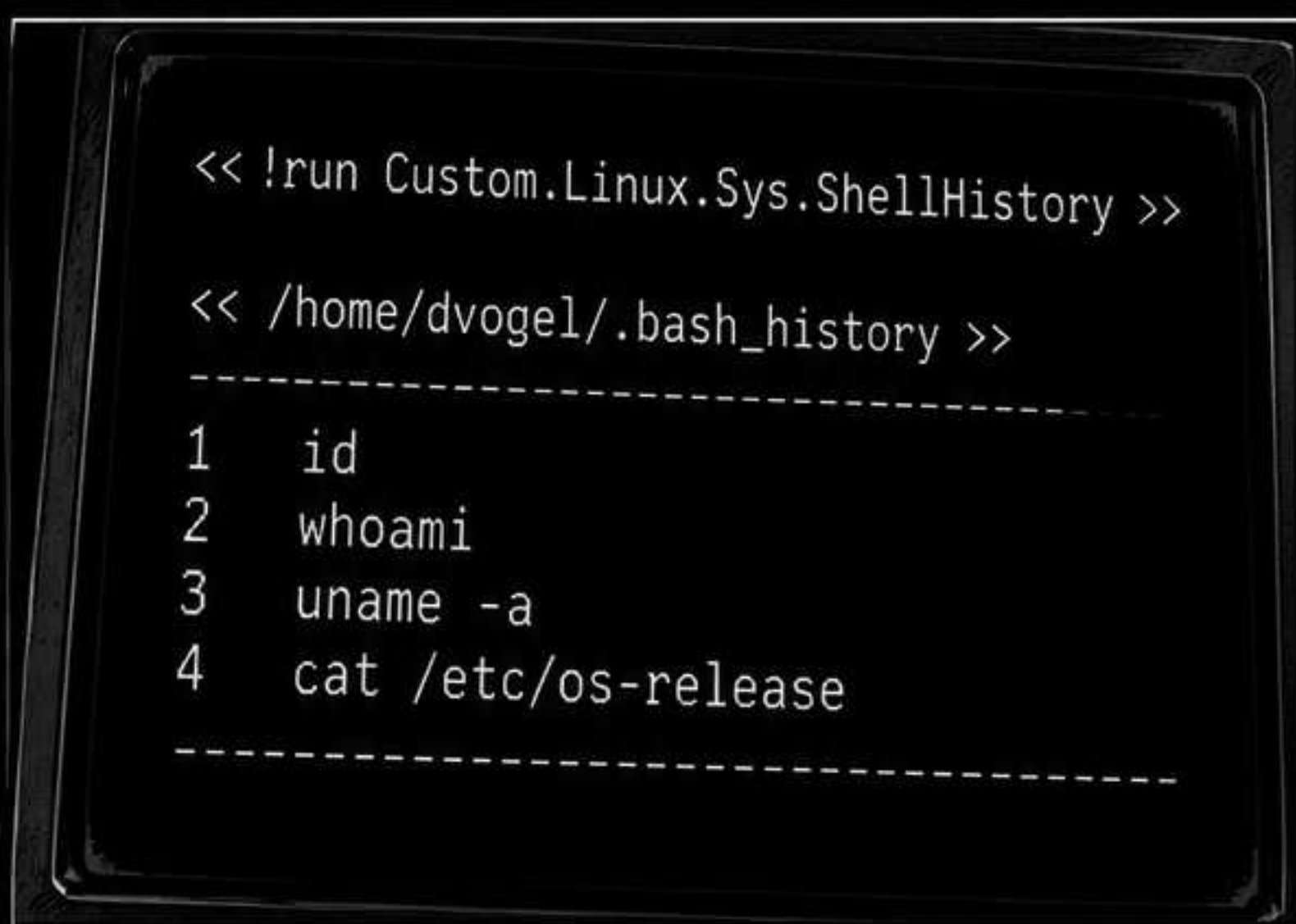
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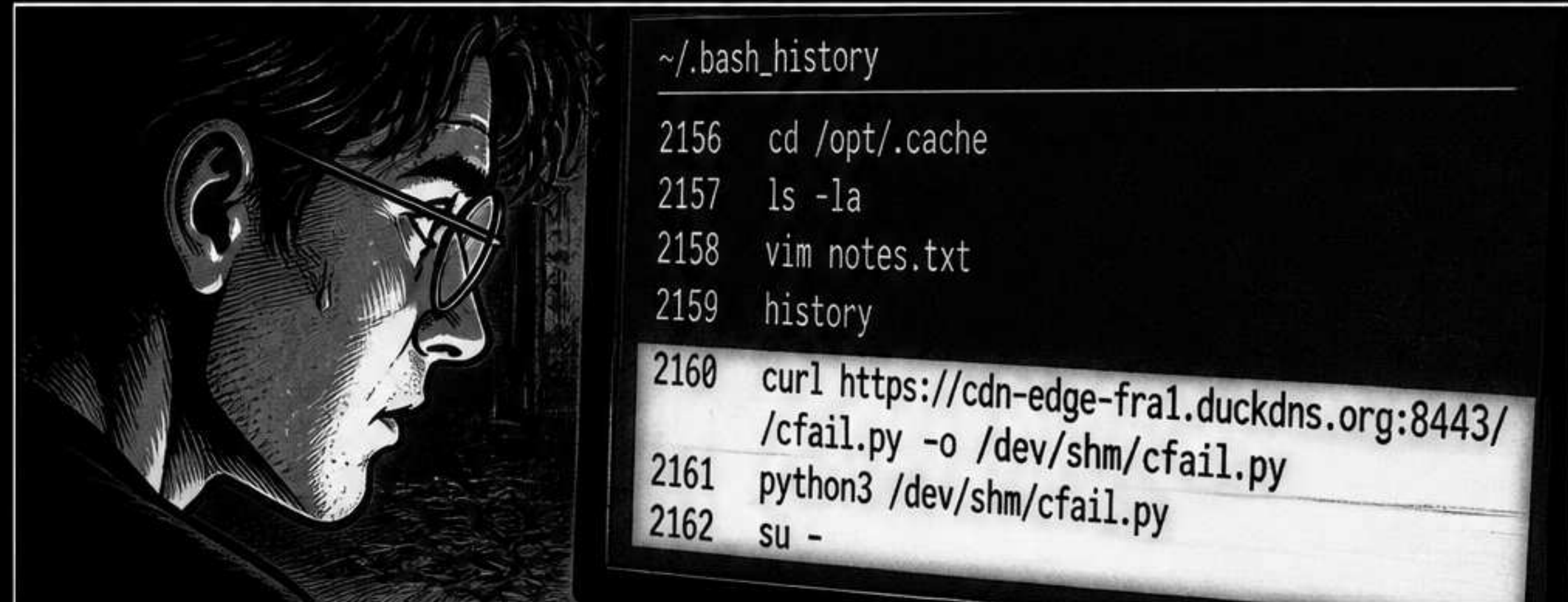
```
<< !run Custom.Linux.Sys.AuthLog >>  
  
<< SuEvents 22:51:42 UTC: su - | InvokedBy: dvogel |  
session opened for root >>  
  
<< UserMgmt 22:52:20 UTC: useradd -m -s /bin/bash svcmon >>  
_  
_
```



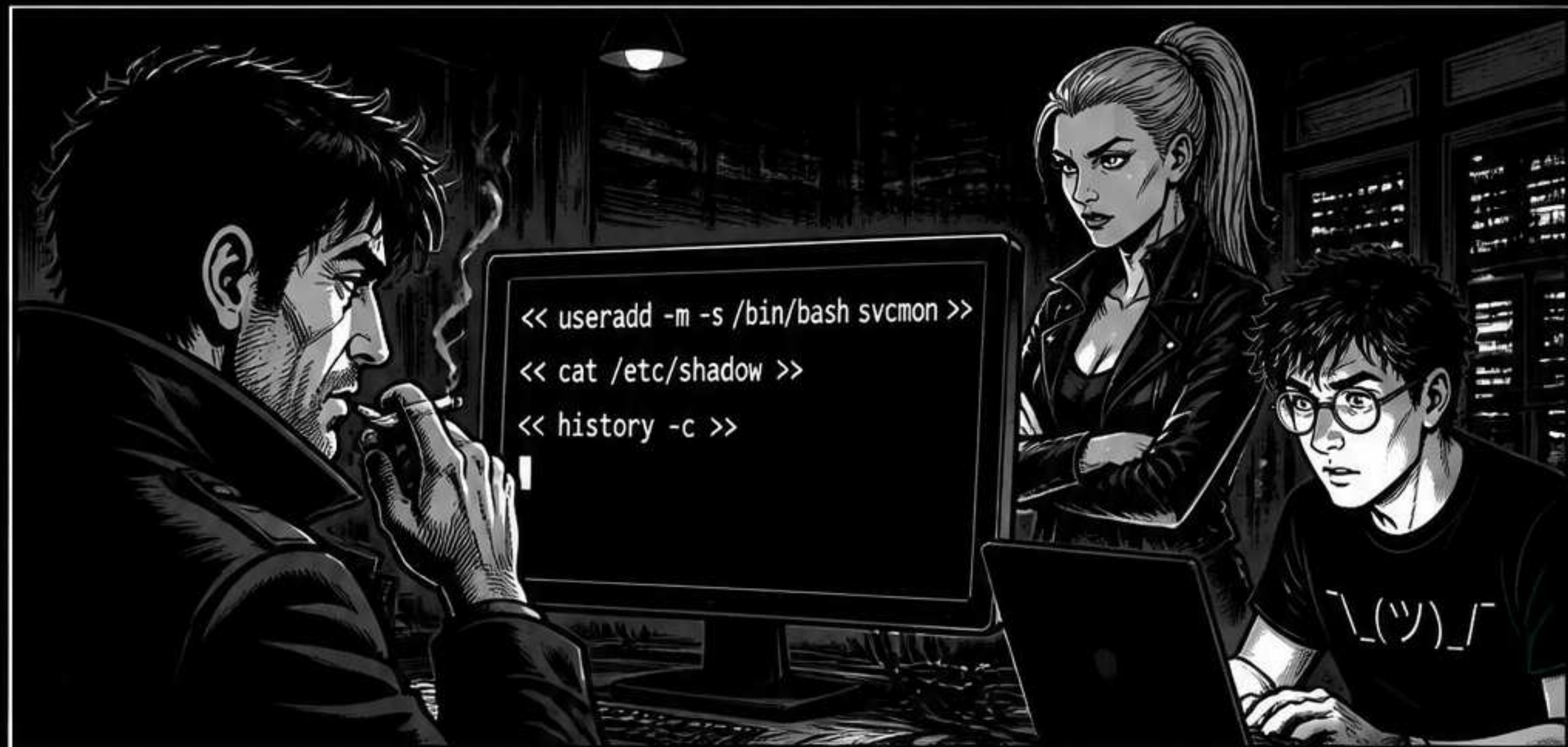
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Backdoor.
Then dumped
shadow.



history -c.
The wipe failed.



It's the last
line he saved.

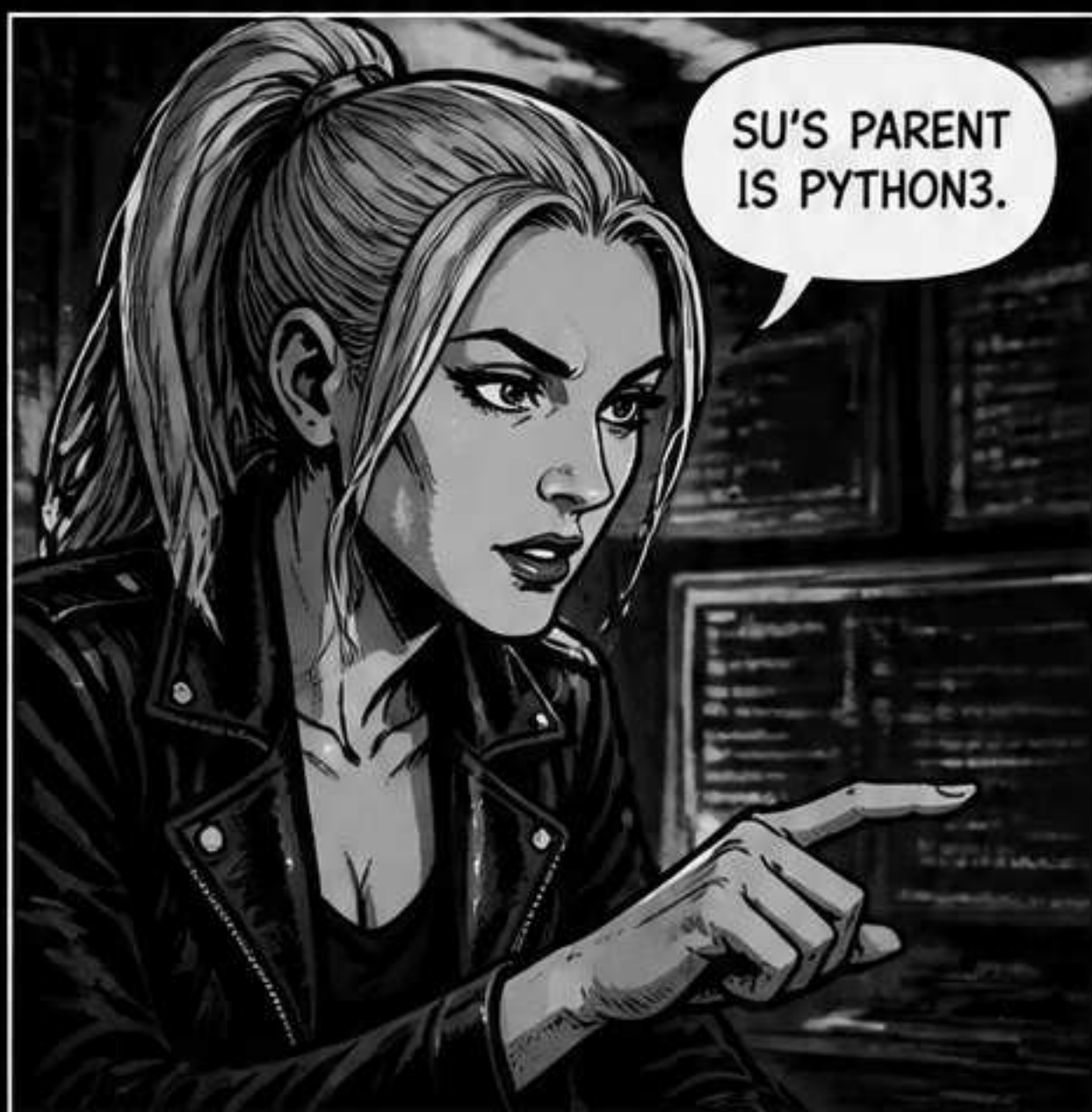
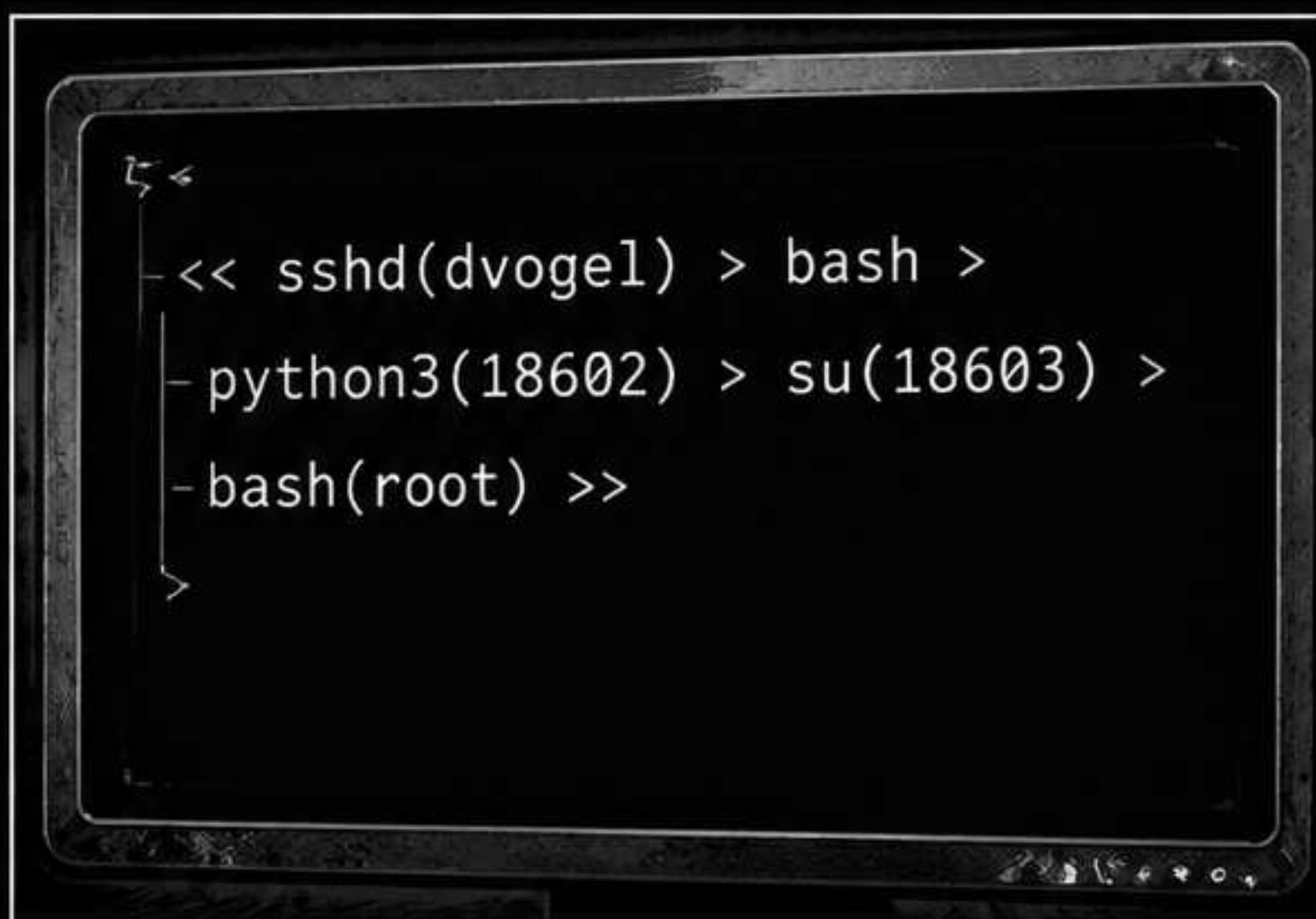


```
<< useradd -m -s /bin/bash svcmon >>
<< cat /etc/shadow >>
root:$6$DwQp4r7J$kLzV7Qe6Y4JXzNfH2yF3j1:19793:0:99999:7:::
svcmon:$6$gF8kLm9N$wQp4r7JkLzV7Qe6Y4JXzNfH2yF3j1:19793:0:99999:7:::
mysql:*:19793:0:99999:7:::
www-data:*:19793:0:99999:7:::
backup:$6$AbCdEf1G$hIjK1LmNoPqRsTuVwXyZ1a2B3c4d5e:19793:0:99999:7:::
dylanlog:$6$XyZ1a2B3$c4d5eFgHiJkLmNoPqRsTuVwXyZ1a2B3:19793:0:99999:7:::
...
<< history -c >>
```

Last
line.



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```
<< !run Custom.Linux.Sys.ProcEnviron >>  
<< pid 18603 su -- no SUDO_USER -- no LD_PRELOAD >>
```

```
Process: su (pid 18603)  
User: root (euid=0)
```

Environment Variables:

```
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin  
TERM=xterm-256color  
LANG=C.UTF-8  
PWD=/root  
LOGNAME=root  
HOME=/root  
SHELL=/bin/bash  
USER=root  
SHLVL=1  
_=/usr/bin/su  
OLDPWD=/home=elf0  
SSH_CONNECTION=10.10.14.27 54312 10.10.10.5 22  
SSH_CLIENT=10.10.14.27 54312 22  
SSH_TTY=/dev/pts/2  
XDG_SESSION_TYPE=tty  
XDG_SESSION_CLASS=user  
XDG_SESSION_ID=2  
XDG_RUNTIME_DIR=/run/user/0  
SYSTEMD_EXEC_PID=18603  
_SYSTEMD_INVOCATION_ID=7e3f2d5c3b0a4f1d9b6c2af1e9d66c  
JOURNAL_STREAM=8:20562  
SUDO_COMMAND=/bin/su  
SUDO_USER=  
SUDO_UID=1001  
SUDO_GID=1001  
PATH_LOADED=/usr/lib/security/pam_env.so  
LESSOPEN=||/usr/bin/lesspipe.sh %s  
LESSCLOSE=/usr/bin/lesspipe.sh %s %s  
MAIL=/var/mail/root  
_=/usr/bin/su
```

Not sudo.
Not LD_PRELOAD.

SOC

So the
binary
itself.

Or the
kernel
beneath it.

~_(_)/~_/
01000110 01000101
</log>



DFIR

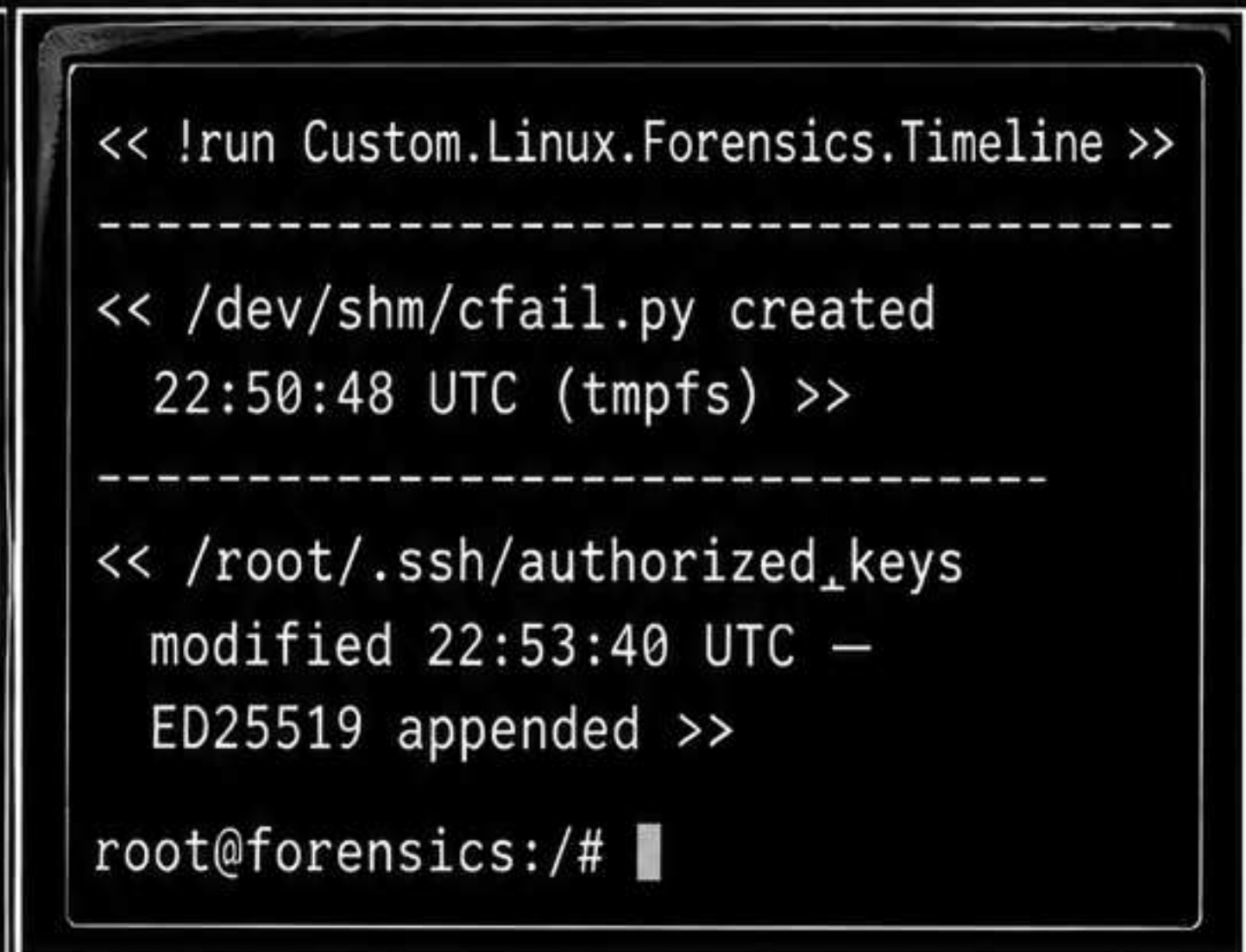
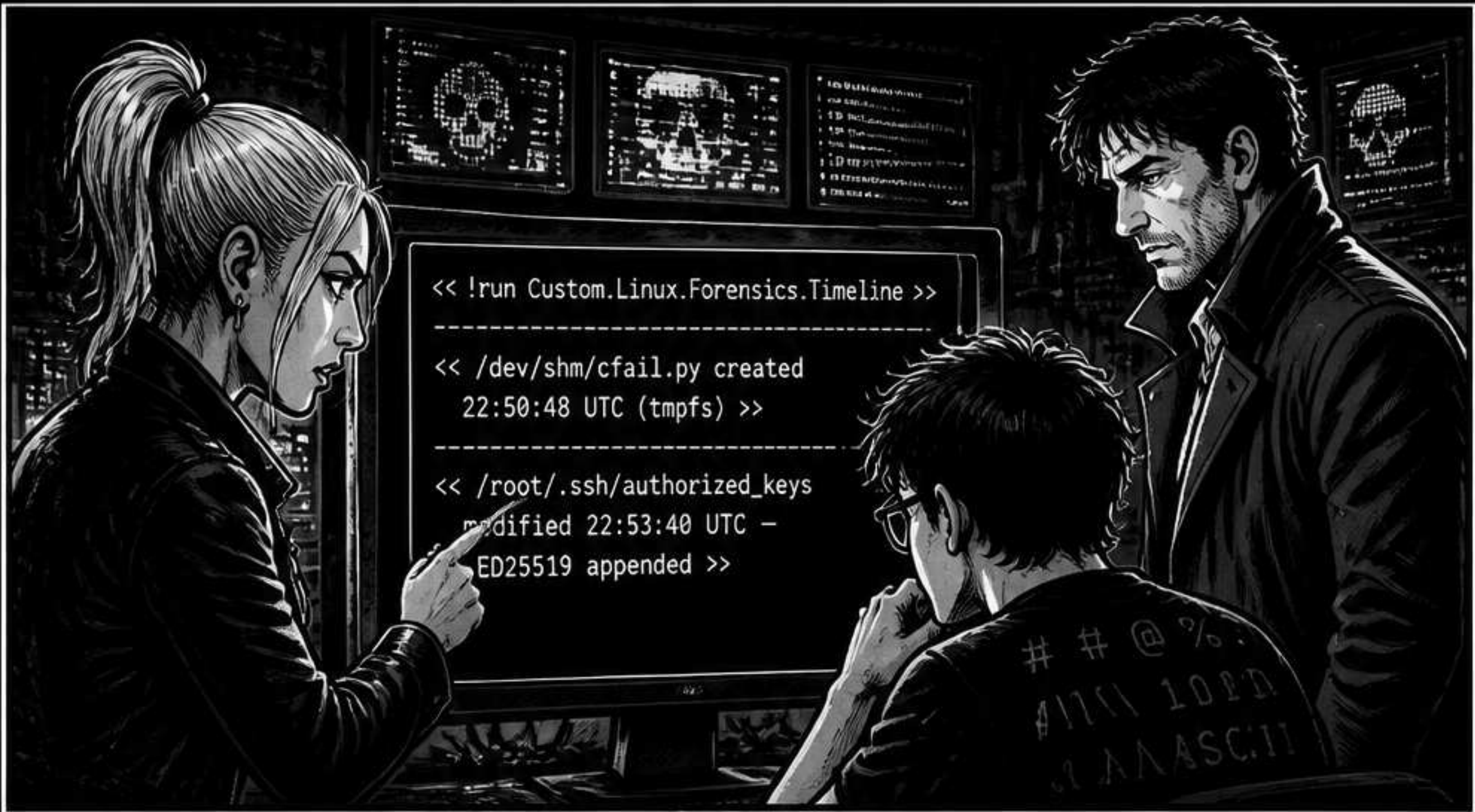
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